

Perception

Consider the situation in Figure 3.1, where you are walking down a street, look around, and have a sense of being lost. You grab your mobile device to search for directions from a navigation application. You unlock the device, launch the application, and press a button to locate yourself on the map. You may have stopped walking to read the map. Alternatively, you may have kept walking but slowed your pace to avoid bumping into other people. *Perception* is a critical capability that made all this possible. You used visual perception to locate the buttons and guide your finger to press them. If you had approached another person while walking, you may have noticed that through your peripheral vision. Beyond vision, you used your tactile sense to guide touches on the device, audition (hearing) to monitor sounds in the background, and vestibular sense to maintain balance.

Perceptual tasks that we may consider simple, like looking for a button on a display, often have plenty going on in the background. A display consists of light-emitting pixels. As you move your eyes, patterns of activations of rod and cone cells occur on the retina, feeding into a perception of regions with colors, shapes, sizes, and orientations. Over many glances (fixations) on the display, you construct a more coherent *percept* of it. This percept has structure. It is not a chaotic galore of colors: some objects are in relation to each other. For example, certain elements appear as being in front of others or belonging together. These percepts ensure that when we search for something, it happens in an orderly and efficient way. We focus our attention not on random elements but on elements that have a higher probability of being the ones we are looking for. This process is affected by both the content of the percept itself and expectations that we have unconsciously learned from previous encounters. After a few glances, you locate the button, read the label, confirm that it is what it is supposed to be, and click it.

This mundane example demonstrates why understanding human perception is fundamental to human–computer interaction (HCI): Perception is the main means of acquiring information about the state of a computer. A user interface (UI) “communicates” to users through perception. Therefore, the way people sense and perceive is essential knowledge to study and design HCI. The example also demonstrates another important property of perception. The design does not fully determine the way we look at the interface; our previous experience and strategies also play a role. To understand how to design perceptually efficient interactive systems, we need to understand how prior experience, attention, and the designed world work together.

More generally, perception is the ability to collect and organize information about the environment through physiological sensory systems. Thus, perception refers not only to the subjective sensory experience of the interface. It also refers to the processes that help us *organize* a representation of the display. Its functioning can be understood via three primary components (Figure 3.2). One component is the *sensory information* that our sensory systems produce. A sensory modality is a biologically specialized system dedicated to a type of transduction. In sensory transduction, a form of physical stimulation is transformed into neural events. For example, when light hits the back of the retina, it triggers a cascade of physiological events through the rod and cone cells, which ultimately contribute to our perceptual experience. Sensory modalities



Figure 3.1 Perception serves a multitude of roles in human–computer interaction. We regulate our actions in interaction via perception. User interfaces communicate their state via perception. In this example, perception has a decisive role in helping us find elements on the display, guide fingers, maintain awareness of the background, and control gait and walking.

commonly used in HCI include vision, hearing (audition), and touch (tactition). There is also experimental research on smelling (olfaction) and tasting (gustation). Moreover, we can rely on the sense of our body, or proprioception. It refers to the knowledge of the position of our limbs in space.

The second component concerns *expectations*. Over years of interacting with the world, we have accumulated experience. Such experience enables the brain to *project* its expectations back to the world. To achieve this, perception utilizes *internal representations* to constantly make guesses on how the world *might* be. Internal representations allow us to deal with the fact that sensory stimulation is relatively poor; in other words, they help us fill in the blanks.

The third component is *attention*. Attention refers to the ability to focus our processing power on a select portion of the full perceptual scene. The decision on what to attend to is strategic: At any given moment, you could be attending to almost anything in your vicinity, but you only attend to one thing at a time. Attention strategies affect what we perceive and are as important to understand as the other two components of perception. If you were left with sensory information only, you would have no way to form a coherent organization of a UI. We need top-down processing to impose organization on sensory data, and we need attention to guide the formation of that organization.

But why is understanding perception important for HCI—is it not something that only psychologists and biologists should care about? As stated in the introduction to this part, understanding people may be put to many uses, such as explaining empirical findings and designing UIs. In the case of perception, we do this by drawing on concepts and models developed in biology, neuroscience, and cognitive sciences. This may help us do the following in HCI:

- Design display technology. A display is a device that presents computer-controlled patterns of physical stimulation to express information in a computer-controlled way (Chapter 25). The paper by Denes et al. [185] is one example of such work. Their study aims to achieve

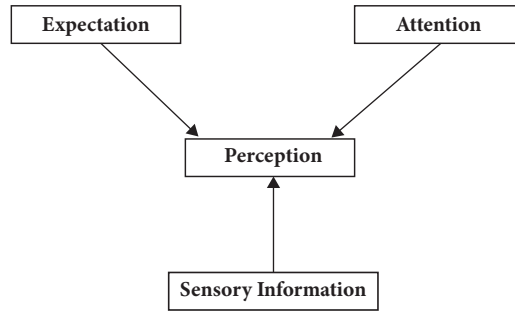


Figure 3.2 Our perceptual experience of the world may appear as veridically (truthfully) reflecting the sensory data we receive. However, it is a representation that our mind actively constructs. On the one hand, perception is affected by our expectations, which are drawn from prior experiences. On the other hand, perception is shaped by how we deploy attention to sample information.

a high number of frames on a VR headset. They do so by exploiting features of the human visual system to reduce the resolution of every other frame. In that way, they can reduce the transmitted data by about 40% with a limited cost to the perceived image quality.

- Explain why people use computers as they do. For instance, Figure 3.1 suggests that the visual search on a UI is influenced by both expectations and the visual features of the page. This is confirmed in eye-tracking studies showing a distinct pattern in search based on users' expectations from previous pages on which the most relevant information is located. We can use such patterns to place information in the most salient places [393].
- Evaluate designs. The understanding presented in this chapter directly underlies guidelines for effective design and may be used to evaluate UIs. For instance, the Gestalt laws for visual perception explain how visual stimuli are organized into unified wholes. This allows us to check if our intended organization of a page is the one that users are likely to perceive (Section 3.3).
- Inform the design of visualizations, interaction techniques, and UIs. For example, laws of contrast perception and the Gestalt laws have been used to algorithmically optimize data visualizations such as scatter plots, enabling people to see structures in complex datasets that would otherwise be difficult to see [540]. In interaction techniques (Chapter 26), we can find optimal ways of providing feedback based on human perception; occasionally, we might even exploit its limitations to make interaction techniques work.

It is hard to design a good UI without understanding the foundations of human perception. Next, we discuss commonly needed sensory modalities in interaction and the key functions of perception in interaction. We then take a closer look at the visual modality, which has a prominent role in present-day UIs. For more information on the anatomical and physiological aspects of perception, see, for instance, [876].

3.1 Sensory modalities

Sensation is a physiological process that produces information about the environment for perception. Sensation feeds perception. From a biological perspective, sensation is about *transduction*. A *sensory system* transforms energy in one form—say, light or physical contact—to electrochemical events in the brain that produce the experience of sensation. This is a complex physiological

process. It is important to understand, on the one hand, how sensation is different from perception and, on the other hand, how the two are interrelated. In the rest of this section, we discuss the differences of sensory systems commonly considered in HCI, how percepts are constructed from sensory inputs, and how they are integrated for multimodal percepts.

3.1.1 Comparing modalities

Human sensory modalities are served by three basic types of transduction [778]:

Mechanosensitivity: Examples of mechanosensitivity are kinaesthesia (sense of own movement), touch, hearing, and equilibrium. Here, the energy of physical contact is transduced, for example, through hair cells in the ear.

Chemosensitivity: Examples of chemosensitivity include gustation and olfaction. Here, chemical properties are transduced, for example, by taste buds on the tongue.

Photosensitivity: Examples include the retina. Here, the stimulation by photons is transduced into electrochemical events in the rod and cone cells.

Another way to look at sensory modalities is based on *what* they sense. *Exteroceptive* modalities sense stimuli outside our body. They include vision, hearing, feeling, smelling, and tasting. *Interoceptive* modalities sense stimuli inside our bodies. They include proprioception and the vestibular sense.

These sensory modalities differ greatly with respect to the properties important for interaction. In HCI, we consider the following differences as important:

1. **Information rate:** How much information can be sensed per unit of time? This is tricky to measure, except for visual and auditory perception.
2. **Parallelism:** How much parallel processing of information can occur? For example, vision is highly parallel; audition can also do parallel processing, though to a lesser extent.
3. **Sensitivity:** The minimum intensity of physical stimulation that a receptor needs to exceed its sensation threshold. Vision has high speed and sensitivity. Such properties can be measured and modeled using psychophysics (see Section 3.2.2).
4. **Receptive field:** The size of the region that produces an integrated feature. It is a measure of the association between neurons and receptors.
5. **Adaptation:** The tuning of outputs to attenuate non-informative signals (see “habituation” below).

These differences are important to understand when considering which modalities to use in a UI. Table 3.1 summarizes three main modalities and their key properties for HCI. Vision is fast and has high bandwidth thanks to parallel processing, and it can be used to communicate information through visual (e.g., color, shape, size), spatial (e.g., layouts), and lexical (e.g., words) features. Audition (hearing) is fast but serial in presentation and can be used to present information via sound (e.g., auditory icons) and voice. Tactition is fast but requires physical contact. It can be used to convey low-dimensional events and information via the amplitude and frequency of vibrations and haptic texture characteristics.

While the three sensory modalities mentioned in Table 3.1 are the most frequently considered in HCI, the human body also has other sensing mechanisms. These include thermal sensing

Table 3.1 Most common sensory modalities studied in HCI and their design implications.

Sensory modality	Key characteristics	Design considerations
Vision	Fast, high bandwidth for parallel processing with a field of view of about 190 degrees (horizontal)	Visual, spatial, and lexical aspects of graphical displays, such as contrast, acuity, use of color, visual primitives, symbols, and text
Hearing	Very fast (about 40 ms faster reaction times than vision), serial presentation, 360-degree detection area	Properties of sound and voice, such as pitch, timbre, melody, and phrasing
Tactition	Fast but limited to areas of physical contact	Properties of haptic stimulation, such as the amplitude and frequency of vibration

and pain, which have been explored as alternative sensory modalities for design. For instance, Wilson et al. [896] developed thermal icons. Like visual icons, thermal icons have a specific thermal profile associated with a meaning.

Paper Example 3.1.1: You have 50 milliseconds to make a good impression

When users see a web page for the first time, how quickly do they form an impression about it? Lindgaard et al. [484] asked participants to view homepages in random order for short durations. They were then asked to rate the pages on different dimensions of design. In one of the studies, the exposure time was just 50 milliseconds. The visual appeal ratings given after just 50 milliseconds were consistent with those given after 500 milliseconds, suggesting that impressions about the quality of a design form very quickly. This calls attention to visual saliency—what grabs users’ attention when they first see a UI.

3.1.2 Constructing percepts from sensory information

Perception must construct an actionable percept based on the sensory data it receives. However, if the input to perception is fleeting activations of receptors (e.g., photons stimulating cells on the retina), how is it able to construct coherent, organized percepts that guide our actions? How can we perceive a button as a button and not just as the numerous sensations that are caused by its LED lights? Two processes are important here: integration and adaptation.

First, let us explain integration. A sensory system consists of receptors, for example, the mechanoreceptors on a fingertip. It also consists of a neural code and brain regions dedicated to *integrating* information over the distributed sensing inputs. Some sensory modalities, like vision, are *topographically projected* in the brain. That is, the topological structure of the receptors is retained in the corresponding receptive fields. In other words, peripheral receptors, such as those on a fingertip, are projected to central neurons in such a way that their *neighborhood relationships* are preserved. This pertains to brain regions dedicated to visual, auditory, and somatosensory modalities. They integrate information over a larger sheet of peripheral receptors. *Convolution* is a special type of topographical projection in visual perception where receptive fields of increasing size form an interconnected hierarchy. In other words, one layer of neurons is dedicated to precise

local information, another layer is dedicated to a higher level of integration, and so on. One benefit of this structure is that it gives perception access to *both* larger homogeneous regions in the visual field and detail in the foveated region. This makes it possible for larger objects to pop out or appear organized. The visual system is specialized in spatiotemporal information; the auditory system integrates information across frequency and time. We exploit such integrative capabilities in UI design, for example, in graphical interface design when considering how to group graphical elements (Chapter 28).

Adaptation is another key process. Environments change, and so must percepts. Have you ever been startled, for example, when an ambient background noise that you were not paying attention to suddenly stopped? A sensory system is adapted to maximize information gain within its sensing limits. In *habituation*, a sensing threshold adapts to continued stimulation. For example, have you ever noticed that if you rest your fingers on a keyboard, you must move them slightly to “sense them again”? This is because when resting your fingers, there is no novel information to perceive. The opposite of habituation is *strategic adaptation*. Perception shows sensitivity to the statistical structure of the environment. For example, we develop attentional patterns of looking at mobile applications [472]. Early research on web pages found an F-shaped pattern for traces of gaze. However, when similar studies were done for mobile applications, a different pattern emerged. Although we tend to look at the top-left corner the most, faces and text are also likely to attract attention. This is a learned adaptive pattern.

3.1.3 Multimodal perception

In most HCI tasks, sensory modalities rarely operate on their own. Everyday interaction *requires* the integration of information not only within a modality but also across modalities. Even when you are pointing with a mouse, four modalities participate: proprioception (feeling the angles of the joints moving), tactition (feeling the palm of the hand move against the surface), audition (hearing the movement), and vision (seeing the hand and the mouse cursor move).

The McGurk effect is a great example of multimodal integration [822]. It is a perceptual phenomenon in which vision *alters* speech. If you look at a face that pronounces the syllable “ga,” but you hear the sound “ba,” your brain might combine the two and create a perception of “da.” Besides specific effects like this, discrepancies between modalities in HCI can distort integration. You may have noticed that in a videoconferencing situation where there is latency between the audio and video channels, it is difficult to follow what the other people say (Figure 3.3).

The McGurk effect is an example of *cross-modal perception*. There are a number of similar cross-modal effects. For example, the *ventriloquist effect* is something that we commonly experience when watching videos with humans speaking [14]. Due to this effect, we perceive the speech as originating from the characters we see on the display even if there is a large distance between the audio speakers and the person speaking on the display. One takeaway from studies of this effect is that vision tends to dominate other sensory modalities.

Another example is *pseudo-haptics*: the creation of a tactile sensation when there is none [683]. In a pseudo-haptic illusion called the hand-displacement illusion, the visual presentation of the user’s hand is dynamically displaced, creating a sense of a force field when no such field exists. For example, if a mouse cursor is slowed down when moving up a mountain on a map, users report a sensation of a resisting force field.

It is important to note that the term *multimodal interaction* has a different meaning in HCI. It refers to the study of novel combinations of technical modalities in input and interaction techniques (Chapter 26). Many modalities can be combined for a given task, for example, speech, gesture, touch, and facial expressions. However, from a biological perspective, even a “regular”



Figure 3.3 Multimodal integration is needed when attending a videoconference. We not only listen to other speakers but also look at their lips and facial gestures when they speak. If the audio signal is delayed, for example, due to network latency, it is difficult to follow what the other person says.

workstation interaction is multimodal. Even when just using a mouse, we engage our tactile, auditory, visual, and proprioceptive sensory systems.

3.2 Elementary functions of perception in HCI

As an applied field, research on HCI pursues the *functional* understanding of perception. HCI has been less interested in the neural and physiological processes that underpin perception and more interested in the properties of perception that affect outcomes and performance in interactive tasks. From a functional perspective, perception is about obtaining the *information* that helps complete some task. Thus, the design of a UI, from a functional perspective, starts with the question: What is *required* of perception for the user to successfully achieve the set goal?

The concept of a *perceptual task* is central to the functional understanding of perception. Some sensory information is available, and the user must make a decision or act in a certain way to reach some goal. Consider the task of finding a link on a web page, keeping a finger on a particular key when playing a game and preparing to respond to an event, or detecting a beep when a new message has arrived. These are all tasks where perception is required, albeit in very different ways. In this section, we look at five elementary perceptual tasks:

Detection: The task of telling whether an event of interest occurs in the environment. For example, was there a link on a page you scrolled quickly through?

Discrimination: The task of telling whether a difference occurs in sensory stimulation. For example, is this object brighter than this other object?

Recognition: The task of categorizing a stimulus. For example, you are playing a game and see something pass your screen quickly—was it an enemy or a friend?

Estimation: The task of estimating a property of an object or event in the environment. For example, how far is an object you see in a virtual reality (VR) environment—could you reach it?

Search: The task of localizing an object of interest. For example, where on the keyboard is the key for the character ≤?

To understand how these tasks are solved by perception, we need to understand both the “hard” and “soft” sides of perception. That is, the hard-wired capabilities of perception that have evolved with us as a species and how, via lifetime learning, they are adaptively controlled and shaped. Our perceptual system has specialized capabilities to support us in tasks that are important for our success. Changes occur via two processes: phylogenetically, via evolution, and ontogenically, via learning. Some physiological enablers of perception, such as the structure of the retina, change slowly over a lifetime. However, many defining physiological characteristics have remained stable for millennia, as they have evolved for tasks important to our species such as finding food, spotting predators, and understanding the facial expressions of our companions. From an evolutionary viewpoint, the time that *Homo sapiens* has spent with computers is too short to have had an impact on phylogeny.

Learning and adaptation to perception occur as a consequence of experience. Consider, for example, learning where to look when looking for a link. We know that the first places where we look reflect the statistical distribution of elements on the screen. On a web page, we first look in the upper left corner; when using a mobile application, we tend to be drawn to faces and large logos [472]. Adaptation can be remarkably fast. For example, when looking for a link, we try to avoid looking at places we have already checked. This “inhibition of return” is important for making visual search efficient.

The rigorous definition of perceptual tasks has been instrumental in advances in understanding perception in HCI. Tasks that can be controlled in experimental settings have enabled the exposition of the underpinning phenomena and the mathematical modeling of perceptual capabilities. In such models, we see the hard limits posed by the capabilities of our sensory organs, as well as the softer, more malleable limits posed by learning.

3.2.1 Detection

In a detection task, the user must decide whether an event has occurred or not. Consider, for example, playing a game and having to decide if a shadowy dark contour in the distance is another player, or a radiologist examining a computer tomography image and having to decide whether cancer is present (Figure 3.5). The study of detection tasks in HCI originates from the study of radar operators in World War II. Radar operators had to examine grainy black-and-green displays and determine whether a blob was an enemy plane. For instance, a blob could also have been a bird or caused by interference in the radar reading. Detection models can inform the design of displays to help people better detect events.

Signal detection theory offers a theory and a model to understand detection performance. It assumes that detection is affected by two components: sensitivity and response bias. In a noisy environment, detection performance is limited by the observer’s ability to discriminate the signal from noise (sensitivity). In Figure 3.4, the HCI researcher wanted to study whether a contrast enhancement algorithm improves the ability of radiologists to detect bone cancer in images. The quality of the X-ray imaging equipment affects the ability of radiologists to detect cancer. Moreover, a novice radiologist, or one with poor eyesight, may be less sensitive to differences.

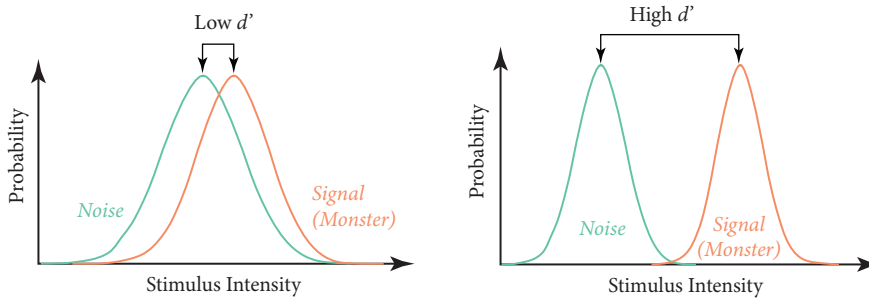


Figure 3.4 You hear a sound during your sleep: is it a monster or just noise? In a hard detection task, the distributions of the two sounds are similar; as they are hard to discriminate, many errors occur. In an easy task, the two sounds are so different that they can be easily discriminated. The d' statistic describes how far apart the two events are.

However, people may not respond simply on the basis of what they sense. They may also strategically adjust their responses, resulting in biased results. Such adjustments may reflect the expected value of failing versus succeeding in a task. For example, radar operation in World War II was not without risks. The consequences of a poor choice were potentially devastating. If an enemy went undetected, people could lose their lives; if noise was incorrectly interpreted as an enemy, time and money would be lost. In this context, the radar operator may be biased toward responding “yes” (enemy) even when somewhat unsure. Let us look at the sensitivity and response bias components more closely.

Will a user perceive something shown on display? The *sensitivity* of an observer depends on the perceived properties of the signal compared to noise. For example, how sensitive is a user to changes in the pitch of an auditory icon? Sensitivity, across modalities, can be empirically measured in a task where a participant is shown signals and non-signals (noise) and asked to report yes/no accordingly. We categorize the responses into four classes: true positive, true negative, false positive, and false negative. The sensitivity can be calculated using this response classification. It is often described using a statistic called d' , which describes the difficulty of detecting a signal, in particular how difficult it is to distinguish it from noise. It can be computed from the observed hit rate (H) and the false alarm rate (FA):

$$d' = z(\text{FA}) - z(H). \quad (3.1)$$

Here, $z(\text{FA})$ and $z(H)$ give the right-tail probabilities of the rates on the normal distribution. We cast the rates to the normal distribution to ensure we can compare sensitivities across different types of tasks.

Intuitively, d' represents an index of how far apart two distributions are:

$$d' = \frac{\mu_{\text{SignalPresent}} - \mu_{\text{SignalAbsent}}}{\sigma}. \quad (3.2)$$

Figure 3.5 shows two examples: one with low and one with high d' . d' provides a way to compare the detection performance across an experimental task. The closer d' is to zero, the less discrimination is possible. When $d' = 0$, the performance is pure chance. The higher the d' , the more sensitive the observer. When $d' = 4.65$, performance is already very high; for example, the hit rate is 0.99, and the false alarm rate is 0.01.

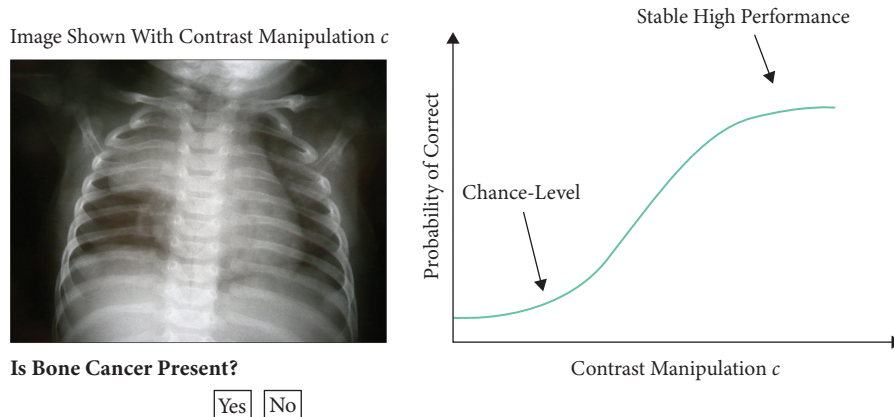


Figure 3.5 A hypothetical psychophysics experiment and a model. Here, an HCI researcher has come up with a contrast manipulation that is expensive to compute but could help radiologists detect bone cancer from images. In the study, radiologists are presented with images and must press yes/no to indicate whether an event (bone cancer) is present or not. Some proportion (e.g., 10%) of the stimuli have the event, while the rest do not. Over multiple trials where the contrast is manipulated, a psychophysical function can be built that tells the odds of detection as a function of contrast. A psychophysical function relates a physical measure to a measure of perceptual ability. Here, the graph shows that stable detection is only reached when the contrast algorithm is applied at a level of $0.8c$ or higher.

Response bias denotes a participant's bias (tendency) toward a particular response. For example, if an enemy in a competitive game shoots me because I failed to shoot first, this can have a high cost. However, if I erroneously shoot an important co-player in my team, this can have an even higher cost. In signal detection theory, the response criterion is denoted with β . A rational player adjusts β to favor a more reserved policy, for example, to avoid shooting co-players. This high criterion means fewer hits but also fewer false alarms. Conversely, a low criterion means more hits but also more false alarms. Response bias can also occur for other reasons. For example, if the "yes" button is easier to reach, a response bias can emerge even if it is not in favor of the player.

3.2.2 Discrimination

In a discrimination task, the user must decide whether a level of stimulation is available or if that level has *changed* due to some reference. The *discrimination threshold* defines the minimum level of stimulation required for sensing. For example, healthy young adults can hear sounds between 20 and about 20,000 Hz. With aging or hearing loss, the discrimination threshold changes. The thresholds for sensing visual contrasts also change. Such individual differences are important to note. All too often, we design interfaces with our peers in mind and forget that many (and in some cases most) users are not the same.

Psychophysics refers to the study of psychological (psycho) responses to physical (physics) events. Its core aim is to measure an observer's ability to distinguish differences in physical stimulations. In computer graphics, psychophysical models are used to optimize rendering,

image processing, haptics, audio, and video. For example, many common video compression algorithms exploit discrimination thresholds. Knowing that a human cannot observe a minute change in color can be exploited in compressing images. In HCI, psychophysics laws have gained recognition among design guidelines [477] and in the design of haptic and other displays.

A psychophysics model is a systematic way to describe people's ability to discriminate stimuli. It relates mathematical physical changes to the perceiver's ability to detect those changes. To obtain data for such models, a participant in a psychophysics experiment is presented with two stimuli, for example, two auditory beeps, and asked to determine whether they are different. The researcher knows if a difference exists and how large it is and uses this information to model the participant's detection capability. By systematically controlling the properties of the baseline stimulus and its difference relative to the other stimuli, this capability can be described for a range of stimulations. In another common task type, the user is presented with a reference stimulus and has to *adjust* another source of stimulus to match it.

A well-known psychophysical model, *Weber's law*, states that equal stimulus ratios produce equal subjective ratios [794]. Weber, a German physician from the nineteenth century, provided different physical stimulations such as weights or intensities of light to people and asked them to state if a difference existed between them. He discovered that there is a relationship between the size of the stimulus and the accuracy of people in identifying differences. Specifically, Weber's law states that the *just noticeable difference* (JND) is proportional to the size of the standard stimulus:

$$\text{JND} = kS, \quad (3.3)$$

where S is the size of the standard and k is a constant called the Weber fraction. S can represent physical quantities such as weights, intensities of light, or pressure. k describes the increase in the standard stimulus required before the observer can make a reliable discrimination. JND thresholds have been charted for perceptual events and qualities relevant to UIs: visual length and area, visual distance, visual velocity, visual flash rate, and duration [794].

One application in HCI has been *time perception*, with applications to the response times of the system and progress bars [755]. For example, the pace of change of a progress bar affects the user's perception of system speed or the perceived duration of loading.

3.2.3 Recognition

In a recognition task, a stimulus must be classified as one out of a number of classes. For example, icons, words, faces, alarm sounds, and so on need to be recognized as a *particular* member of a set. "This is the icon of Excel." All other things being equal, the larger the set, the harder the recognition task. It matters whether you have 12 or 100 icons on a display. Discrimination becomes harder as the number of candidates increases.

People are surprisingly good at recognizing faces and objects they have seen before. In controlled studies, participants can accurately recognize hundreds of objects that were presented to them previously [617]. Performance in a recognition task can be measured by precision and recall. *Precision* is the number of true positives divided by the sum of true and false positives. *Recall* is the number of true positives divided by the sum of true positives and false negatives.¹

¹ Not to be confused with memory recall; see Chapter 5.

Human ability in visual recognition is leveraged in the direct manipulation paradigm of graphical user interfaces (GUIs; Chapter 28). It is important for training better machine learning algorithms. Humans are often asked to classify data that are used to train algorithms. However, people do not always agree on what the “true” classes are. Categories that show disagreement among human observers are problematic for machine learning because they add noise to the training data. HCI researchers can help build UIs that improve recognition accuracy and help annotators solve conflicts effectively.

3.2.4 Estimation

Estimation is the task of assessing a property of an object from stimulus information. Users may be estimating, for example, the size of a 3D object in a VR environment or the color of a product on an e-commerce site. Here, they need to assign the estimated value to some property. For example, they may estimate how far away another car is based on what is shown on the display of a self-driving vehicle.

According to *cue integration theory*, our prior experiences affect our estimations. Over time, users become more sensitive to how likely different properties are [231]. The theory assumes that people consider all information available to them optimally. For example, when trying to respond to a serve in a virtual tennis game, we need to estimate the ball’s speed. To this end, we have two types of cues at our disposal: (1) the sound we hear when the ball was hit by the opponent and (2) the motion of the ball as we see it move on the display. The two cues provide unique and complementary information.

The theory suggests that we integrate cue-specific percepts into one estimation of the ball’s velocity. The idea of optimality is that we make an optimal estimate based on the observed data (the two cues). In new tasks, where users have little exposure to the cues, they would struggle to make a good estimate. However, with more time, their performance improves and is better described by the theory. Despite the optimistic assumption of optimality, the theory has turned out to be accurate in fast-paced perceptual tasks in HCI, for example, in tasks where a moving target needs to be selected [468].

3.2.5 Search

In search tasks, the location in space of some object must be determined. For example, a user may need to localize an icon on a display or a command in a menu. In auditory search, the task is to localize the source of a given sound. We talk about visual search as a case below in Section 3.3.5.

3.3 Visual perception and attention

Visual perception refers to perception through the sensing of stimulation by light. It is a sensory modality of prime importance in HCI. Vision offers a rich and efficient way of conveying information to a user. First, visual perception is fast. Unlike olfaction, visual sensory experience is fast, formed within milliseconds from the onset of the stimulating event. However, constructing a more comprehensive percept, which requires several eye movements, can take seconds when the display is complex.



Figure 3.6 Wear's windows of visibility can be used to understand how viewing conditions affect perceptual tasks in the wild. For example, consider wanting to glance at a smartwatch while outdoors. Using the windows of visibility, assess the situation: What are the key limits to the user's visual performance? Photo used under Creative Commons Zero.

Second, it offers unparalleled parallel processing. It benefits massively from parallel processing over the entire field of perception. Unlike the sense of touch, with visual perception, no physical contact is required with the perceived objects. The human visual system is proposed to involve three neuroanatomically distinct pathways [662]:

What: The ventral pathway encodes the identity of visual objects, such as tools, faces, or animals.

In the case of GUIs, it recognizes the types of elements and objects on the display.

Where and how: The dorsal pathway encodes the locations of visual objects and actions related to these objects. In the case of GUIs, it registers which elements can be clicked and tracks moving objects.

Who: A more recent discovery, the superior temporal sulcus specializes in dynamic social processing. It processes the actions of moving objects and bodies, such as their expressions, gaze, intentions, and moods. In the case of GUIs, it is employed in contexts such as VR, games, and videoconferencing.

Third, the visual system offers a hierarchical and highly interconnected structure. *Visual primitives*, such as the shape, size, orientation, color, and motion of objects, are extracted in massively parallel processing in low-level areas. The low-level areas feed into high-level areas that recognize objects and encode their identities. These, in turn, are projected back to the low-level areas. This structure realizes the top-down and bottom-up processing discussed in the previous section.

In the following section, we discuss the limits of the visual system and then cover the basics of eye movement, perceptual organization, and visual attention.

3.3.1 Limits to the human visual system

Take a look at Figure 3.6. The photo shows an example of an extreme environment encountered when using computers. Even if the face of the smartwatch may look fine when printed on the pages of this book, the real-world experience would be challenging. Before reading on, think about the following: What is it exactly that makes the case challenging for the user?

Ware [876] summarized the key limits of the visual system under the concept of *windows of visibility*. A “window” is a metaphor to describe the physiological limits posed to visual perception. Here, we describe four windows that are important in HCI: (1) the visible spectrum of light, (2) the field of view, (3) contrast, and (4) foveated vision. First, we can only perceive a limited range in the spectrum of light, from about 380 to 780 nm. Our rod and cone cells permit only three chroma of color (trichromaticity). This sets limits to the range of light that displays need to cover.

Second, our field of view is limited to about 190 degrees horizontally and 125 degrees vertically. This sets limits for how large displays can be and how and where they must be located in relation to our eyes.

Third, our perception of detail is limited. *Contrast* refers to the differences in luminance and color that make something in the field of view stand out from the rest. *Contrast sensitivity* refers to our ability to distinguish between levels of contrast. The contrast sensitivity function is a standard way of expressing this as a function of spatial cycles per degree. The function shows thresholds for contrast perception in a typical viewer. At first, contrast sensitivity is low because the information density is low. The contrast sensitivity is highest in the middle when the cycles-per-degree value matches the properties of the human visual system. As the cycles-per-degree value increases, the sensitivity decreases, as the system is not able to discriminate between cycles.

Fourth, the retinal image is limited and very non-uniform. It loses accuracy at the periphery. Vision is physiologically divided into two regions:

Foveal vision: It refers to a narrow but precise, high-fidelity perception around the point on which you fixate. This area spans only 2 degrees of angle, roughly corresponding to the size of your thumb held out at arm’s length.

Peripheral vision: It refers to the rest of the field of view, which has eccentrically decreasing fidelity. The farther away from the foveal area, the lower the fidelity.

In the peripheral region, vision retains some access to visual primitives but not so much to objects. The visual primitives of human visual perception are color, shape, size, and depth. *Drop-off functions* describe how information is lost going from the foveal to the peripheral regions. An example is given in Figure 3.7.

In the case of the smartwatch (Figure 3.6), which windows are relevant? The answer is at least three of them. First, the watch may reside outside the field of view depending on the posture of the user. To view it, the smartwatch must be brought closer to where the eyes are operating. Second, even then, the smartwatch is likely to be foveally viewed because the eyes need to attend to the shoelaces. To attract the user’s attention, the smartwatch would need to use highly salient visuals (e.g., salient colors, motion). Third, given that the watch face is small and far away, all its text is beyond human contrast sensitivity. Typically, users bring the watch closer to their eyes for reading.

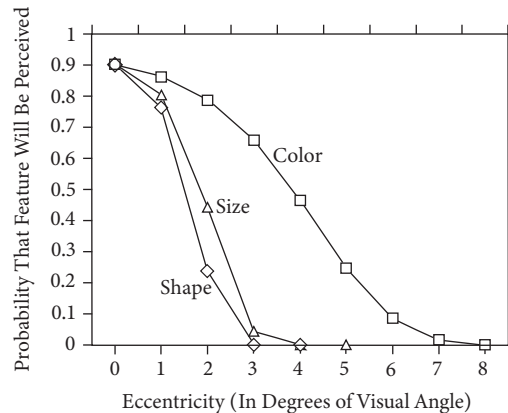


Figure 3.7 Peripheral drop-off functions. The farther a feature on display is from the foveal region (the area of high acuity), the less probable it is that it is available for perception [413]. Note that different perceptual tasks have different drop-off curves.

Visual impairments: Visual impairments that limit a person's ability to view computer displays are surprisingly prevalent. Red–green color deficiency, for example, is about 8% prevalent in European Caucasian males and 0.4% in females, and between 4% and 6.5% in Chinese and Japanese males [73]. Visual impairments include a wide range of physiological and neurological problems that cause color blindness, low-level vision, blindness, and deficiencies in controlling eye movements.

Color blindness, or color deficiency, manifests itself as difficulties in perceiving differences between colors, the brightness of colors, or different shades of colors. People with color blindness are often unable to distinguish certain colors, most commonly green–red and blue–yellow. Humans perceive color using cone cells in the retina that detect either red, green, or blue light. The outputs from these cone cells are then passed to the brain to produce the impression of a particular color. When cone cells do not function as expected, or are absent, this gives rise to color blindness. Color blindness covers a spectrum from mild color blindness that may be affected by, for example, dim lighting to severe color blindness that results in perceiving the world in shades of gray only. Color blindness is usually stable and affects both eyes. There is no treatment for people born with color blindness. However, various visual aids are available, such as apps that allow the user to take a photo and point anywhere on the photo to get the app to identify the color at that point. More generally, the effects of color blindness can be mitigated in GUI design by carefully choosing how information is encoded using colors. In Chapter 25, we talk about HCI research automating color decisions to help users with color perception deficiencies.

3.3.2 Eye movements

The human oculomotor system controls eye movements. It offers three main modes of controlling where we look:

Fixations: They encode information about the visual scene consisting of multiple micro-fixations, each lasting a few tens of milliseconds. What eye trackers do is cluster micro-fixations into

fixations of 200–400 ms. The way we pick information during a fixation is task-dependent. For example, when reading text versus searching for an icon, we pick different visual information (e.g., color, text, shapes, orientations, motion).

Saccades: Saccades move the gaze point in ballistic leaps that are not perceived during the scene. They are ballistic in the sense that the target of the saccade is not changed after the onset of the movement. They are also blind in the sense that no information is sampled during a saccade.

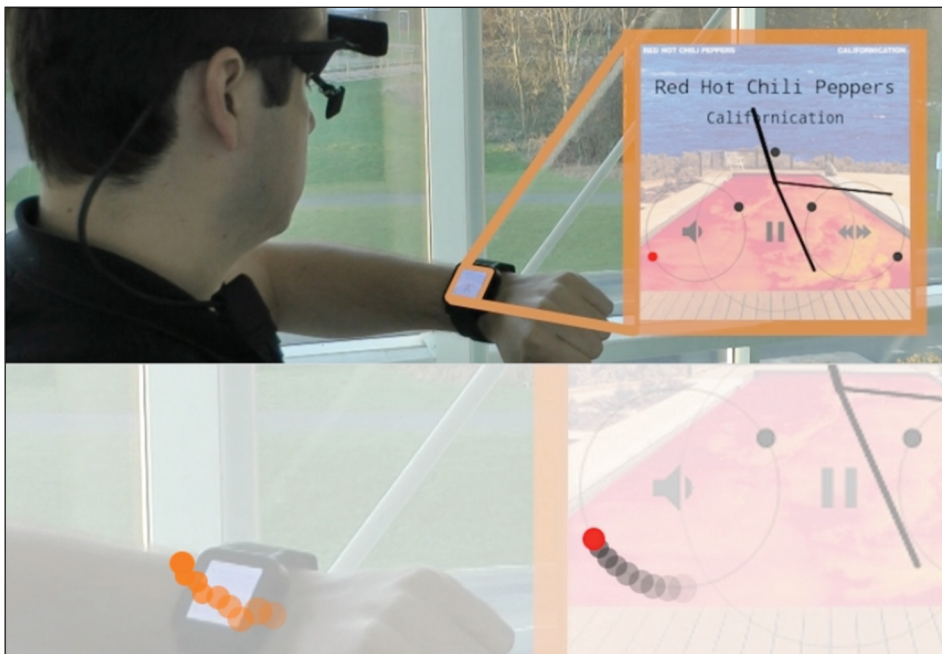
Smooth pursuit: It is used for following moving targets, such as when following an animated character moving on the display. No saccading occurs. An example of an innovative application in HCI is shown in Paper Example 3.3.1.

The speed and accuracy of eye movements can be modeled mathematically; see Paper Example 3.3.2.

Fixations are the main means for the perceptual system to sample the visual environment. Every micro-fixation can gain progressively more information. As described above, this amount decreases *eccentrically*; that is, the farther an object is from the foveal region, the less information we gain about it in a fixation.

This process explains how eyes move when reading text. When reading, the visual system gains information about the fixated word and the following word [155]. The control of the eyes is affected by lexical processing. In turn, lexical processing is affected by three properties: the frequency, length, and contextual predictability of the word. For example, if a phrase is hard to understand, we may need to fixate on words we had already fixated on. When a phrase is easy to grasp or familiar, we may not need to look at every word and may be able to skip some words.

Paper Example 3.3.1: Exploiting smooth pursuit for input



Orbits is a selection technique that exploits smooth pursuit as an input modality [232]. Gaze-based selection normally uses dwell time to detect the object the user wants to select. For example, if one looks for one second at an icon, the icon is selected. The idea behind Orbits is matching. The user needs to move their eyes to match the motion or trajectory indicated by a to-be-selected widget. For example, the user can raise the volume of a player on the smartwatch by following the trajectory of a moving circle.

Paper Example 3.3.2: Speed and accuracy of discrete eye movements when viewing displays

Mathematical models of eye movements are important tools in HCI. They can be used to predict the time costs of using complex displays. Such costs depend on the layout (e.g., distances and sizes of targets) and the expertise of the user. Salvucci [724] proposed the eye movements and movement of attention (EMMA) model to better predict the performance of eye movements in HCI. It consists of equations for predicting the time spent encoding fixated objects as well as the movement time and accuracy of saccades.

First, when fixating on an object, the time needed to encode it, T_e , is a function of the amount of experience with the object. Objects we have encountered more often take less time to encode, and vice versa. This relationship is captured as follows:

$$T_e = K \cdot [-\log(f)] \cdot e^{k\epsilon}, \quad (3.4)$$

where K and k are constants and f is the frequency of the object. ϵ is the eccentricity, measured as the distance of the target from the current eye fixation (in degrees). When the eccentricity is high, to account for its high cost, the visual system may initiate a saccade to get closer to the target. According to the EMMA model, an object (e.g., a word) can be skipped if sufficient information has been encoded during a fixation.

The duration of a saccade, T_s , depends on the distance to the object D , measured in degrees:

$$T_s = t_{\text{prep}} + t_{\text{exec}} + D \cdot t_{\text{sacc}}, \quad (3.5)$$

where t_{prep} , t_{exec} , and t_{sacc} are constants related to the human visual system, specifically to movement preparation and movement execution.

An important property of eye movements is that they are noisy and therefore inaccurate. The actual landing point often differs from the intended landing point. This error can be modeled as signal-dependent noise. The inaccuracy follows a normal distribution with a standard deviation of σ_V times the distance D . That is, the greater the distance, the greater the noise [724].

When applying the EMMA model, the constants need to be calibrated to the particular viewing conditions. Previous applications provide indications for calibrating the constants. Note that the model does not predict the duration of fixations but the encoding time. Encoding can happen over multiple fixations.

Another feature of attention is that we do not always need to move our gaze to encode information. When the encoding time is lower than the movement preparation time, the target can

be encoded without moving the eyes. These eye movements are called *covert* movements rather than *overt* movements. Covert movements explain many of the variations we see in the duration of fixation.

3.3.3 Perceptual organization

When you look at a graphical display, it nominally consists of just pixels on a two-dimensional array. How is perception able to make sense of it? Consider the example in Figure 3.8. Information appears to be organized into regions defined by the desktop, the menu bar, and the window. Some regions, like the window, are standing in front of other regions, like the wallpaper. Some elements belong together and form regions. For example, the list of recommended apps consists of a group of icons that we tend to perceive as different from other application icons.

Figure/ground perception refers to the organization of visual experience in a visuospatial hierarchy. That is, some objects belong to an object in front (a figure) while others belong to objects in the background (the ground). Graphical interfaces utilize the figure/ground perception to show a *display hierarchy*, or screen hierarchy. It defines to what element or whole another element belongs: this belongs to that, and that is different from this. We use cues such as shadows, occlusion (an object covering another), and different sizes to parse the display. When binocular stereo displays are used, stereoscopic depth cues can also help with forming the figure/ground perception.

The perception of organization decomposes a display into *regions*. For example, look at the GUI in Figure 3.8. What you most likely experience is a layout consisting of a few main regions, which in turn consist of elements with text or images. How do we achieve this? If you think about it, there are actually many ways to parse a layout into regions. Two elements A and B are associated

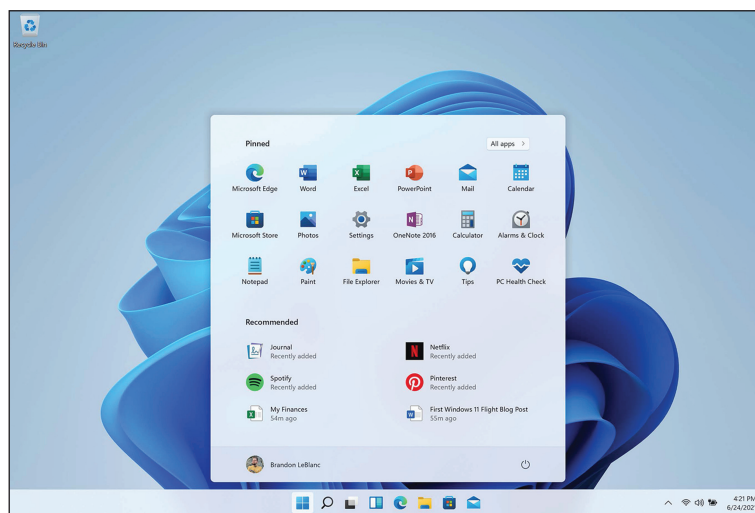


Figure 3.8 Perceptual organization refers to (1) the division of elements into figure versus ground and (2) their grouping into coherent regions. In this example, the application window (figure) is clearly in front of the desktop wallpaper (ground). There are also many visual groups on the display. For example, the region of recommended apps forms a distinct region from the rest of what is presented in the window. Similarly, the icons on the horizontal taskbar form a visual group.

by their similarity in color, distance, and alignment. Elements B and C have the same qualities but different values. How do we determine that two elements belong to the same group and not to some other group?

Visual grouping refers to the tendency of people looking at elements on a display to organize them in visual groups. The *Gestalt laws* originate from a German branch of psychology from the early twentieth century. The goal of Gestalt psychology was to understand what constitutes a whole. Wertheimer studied the visual factors that cause the perception of elements as belonging together on a complex display. He identified seven laws that are now called the Gestalt laws. These laws have endured well [864]. Below, we present four commonly used principles of visual grouping and a related example (Figure 3.9):

Proximity: The closer some elements are to each other and the farther they are from other elements, the stronger they are grouped together.

Common area: Elements that are located in the same closed region are grouped together.

Similarity: Elements that are similar (e.g., in color, size, or orientation) are grouped together.

Continuation: Elements that are connected by the continuation of flow are grouped together.

Brumby and Zhuang [104] found that when menu groups are marked using visual grouping cues, items are found more quickly. However, this strategy only works when there are fewer, larger groups. Having multiple small groups dilutes the facilitatory effect of visual grouping.

Although the Gestalt laws are called *laws*, they are less like scientific laws in, say, physics and more like heuristics to understand certain tendencies in how perception works. The laws are popular among designers because they offer a good way to estimate how users perceive structure in graphical interfaces and layouts. They help us understand how to present the hierarchy of a UI in a way that users find easy to match. They are also part of design education and are frequently used by interaction designers. However, they ignore the fact that perception is an active and adaptive process: Depending on the task and their beliefs, users can organize the display in different ways. We return to discussing active perception at the end of this chapter.

Ungrouped	Proximity	Common area	Similarity	Continuation
Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp	Email Word Excel Powerpoint Drive Mastodon Facebook Snap Instagram WhatsApp

Figure 3.9 Principles of visual grouping. On the left, the figure shows an ungrouped menu design. The other columns show how the principles can be utilized to visually communicate which menu items belong together.

3.3.4 Visual attention

Attention is the focusing of perceptual processing on a region or object in the perceptual field. *Visual attention* is traditionally considered to consist of three processes:

Selective attention: The ability to direct attention to a desired object or location.

Vigilance: The ability to sustain attention on something for a long period.

Divided attention: The ability to split attention between one or more objects, locations, or tasks.

Selective attention refers to the deliberate refocusing of attention to another object or region in the perceptual field. *Overt attention* refers to the movement of eyes to fixate on something. However, we can also sample information without moving our eyes. *Covert attention* is the processing of extra-foveal information without moving the eyes. When reading displays, both covert and overt attention are utilized. Eye trackers are instruments used for measuring overt attention. The limit of how long we can hold our attention on a task or object is called *vigilance*.

Divided attention refers to the division of perceptual processing between multiple objects or regions. For example, *situation awareness* is the ability to keep track of events and objects in a dynamic setting. Consider driving a car through a busy intersection, or an air traffic controller keeping track of planes in the airspace of an airport. Such tasks require us to continuously shift our attention between multiple objects and update our mental representations of their properties. For example, when walking down a busy street, our visual attention on a mobile device is fragmented; the average glance lasts just 3–4 seconds [626]. However, if we walk down a less crowded street, we can maintain our gaze on the device for longer periods.

3.3.5 Visual search

Visual search is a common task with graphical interfaces involving selective attention. The goal is to find an element on the display, for example, an icon or a label of interest. This task poses a control problem for the visual system, which must decide where to place the next fixation. How does it do this? Three things interact to guide the eyes when searching for something: (1) the visual features of the display, (2) learning, and (3) strategic decisions on *how* to search.

First, the display consists of a spatial distribution of visual features. Compare, for example, an application menu and a mobile app in terms of how colors, sizes, and shapes are distributed. Depending on what you are looking at (your current fixation), these features are differentially available to the visual system. If you look at the top-right corner of a mobile app screen, different parts of the visual feature distribution are available compared to if you were looking at the center of the screen.

In *disjunctive search*, or feature search, the visual search is based on so-called visual primitives such as color, shape, size, curvature, orientation, and motion. When the target is unique in any of these features, the search is fast (typically < 100 ms). For example, if you are looking for a green icon on a display that does not have other green elements, you can spot it almost instantaneously if it is large enough. Feature search is fast because such features can be spotted peripherally by the perceptual system. By contrast, in *conjunctive search*, the target cannot be singled out with a single feature: We need to look for a combination of features. For example, if there are multiple green elements, the only way to know which one is the icon you are looking for is to look at each icon and its label. In such cases, the search cannot be “solved” with peripheral processing: It requires fixating on candidate items to pick up the relevant information. This type of search is much slower. In the design of GUIs, elements that should be quickly found can be designed to be found via disjunctive search (Chapter 28).

The *feature integration theory* of Treisman and Gelade [829] explains why this difference emerges (Figure 3.10). The low-level visual features of a stimulus environment are directly represented in what she calls *feature maps*. Attentional spotlight, or information collected via fixations, on the other hand, is needed to find *combinations* of these features. This combined map integrates low-level features; it maps objects (and their features) to their locations. In other words, selective attention is needed to bind features to recognizable objects that can be further processed.

What this model does not account for is the fact that there is less information in the peripheral view than in the foveal view. Kieras and Hornof [413] proposed a model called *active vision*. It emphasizes that people choose where to look based on the description of the target and the visual features available via peripheral vision. However, peripheral information is compromised: The farther away eccentrically an object is from the current fixation point, the more it is compromised. The visual system has to deal with uncertainty about the regions of the display it is *not* attending to immediately; it must “gamble” with its priorities and look at those regions later. According to the theory, it does this only after checking candidate items close to the foveal region. The model is dubbed “active” because users are not passively drawn to visual features; their choice of the next fixation points depends on what they are looking for and what they presently see.

However, these models do not account for the effect of *learning*. If you have already seen an item a few times, it is easier to locate it. In menu systems, for example, a user who has experienced the design a few hundred times can fixate on it virtually immediately when looking for it [127]. The *Guided Search* model of Wolfe [910] attempts to explain this (see Figure 3.11). It proposes that decisions on where to fixate are guided by *priority maps*. A priority map combines bottom-up and top-down information to guide the user on where to look next. Depending on how strong the

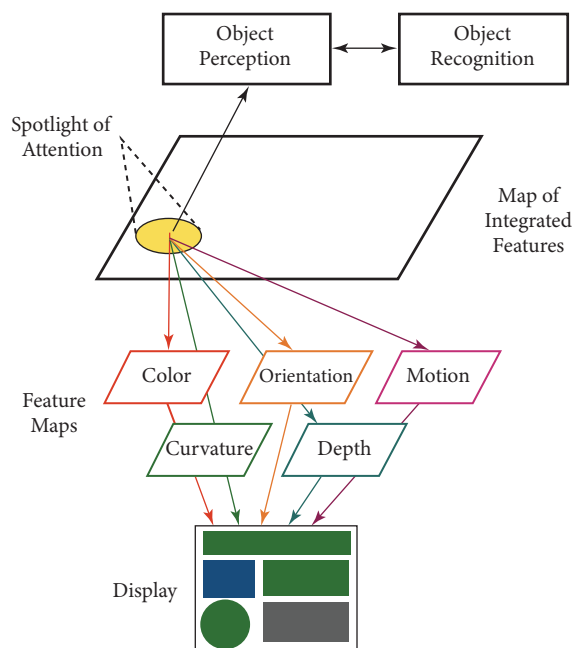


Figure 3.10 Feature integration theory explains why conjunctive search is slower than feature search. Feature search can be carried out using parallel processing of the peripheral vision, whereas conjunctive search requires users to fixate on the region. User interface designers can exploit these differences.

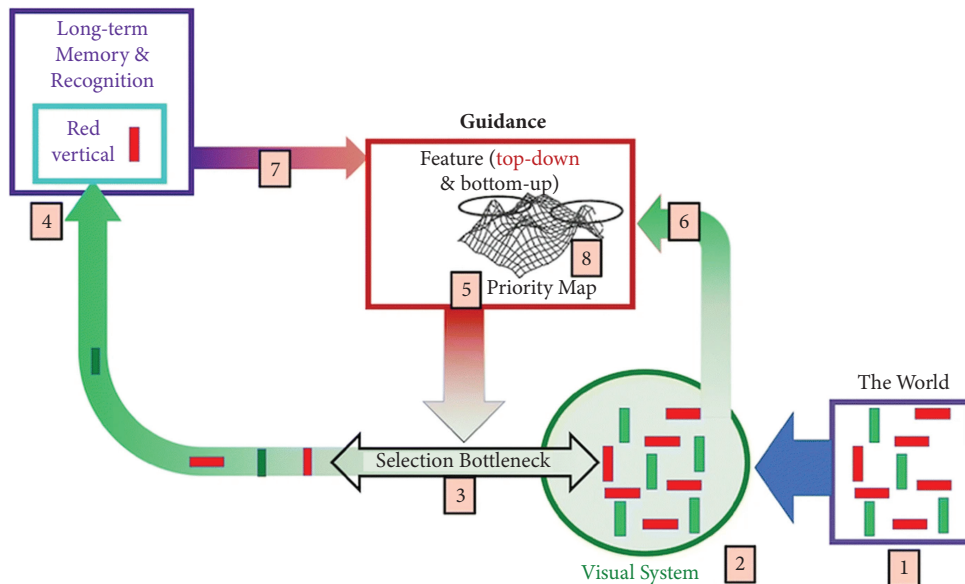


Figure 3.11 A model of visual search by Wolfe [910]. The model suggests that both the long-term memory of locations (top-down) and the perceived features (bottom-up) contribute to the selection of where to place a fixation. This takes place via what is called the *priority map* where this information is combined. This model (Guided Search 2.0) has been updated to also consider scene composition, lexical and semantic features, and task rewards.

long-term memory is, it differentially contributes to the map and therefore to guidance. Users who have seen a UI a few times have positional memory of where objects are located on it, especially those that are often searched for. With more repetition, the locations of elements will be remembered and do not need to be actively rediscovered every time. However, if the display changes, the value of long-term memory is reduced or completely lost, and the user needs to rely on bottom-up guidance again. Wolfe's model has been updated a few times. The version shown in the figure is 2.0. Since this version, the model also covers the effect of task rewards (Chapter 21) as well as scene syntax and semantics.

Users often explore displays to form an actionable priority map. For example, when seeing a new ticket vending machine, should one try to directly find the entry point to the display (exploit) or first become familiar with the overall layout of the system (explore)? We often explore briefly and then search. In such cases, we must decide whether *enough* is known about the scene to find the target of interest, or if more information needs to be collected.

3.3.6 Visual saliency

Visual saliency is an application of visual attention modeling that has become popular recently. Saliency refers to the probability with which a graphical element can attract visual attention during the first few seconds of viewing a display (Figure 3.12). Saliency depends on the visual properties of the target and those of the rest of the display. It also depends on the tendencies learned from repeated exposure.

Attention, generally, is drawn to *visually unique elements*. Through their different color and bold font, these words stand out relative to the rest of this paragraph. Generally, regions and elements

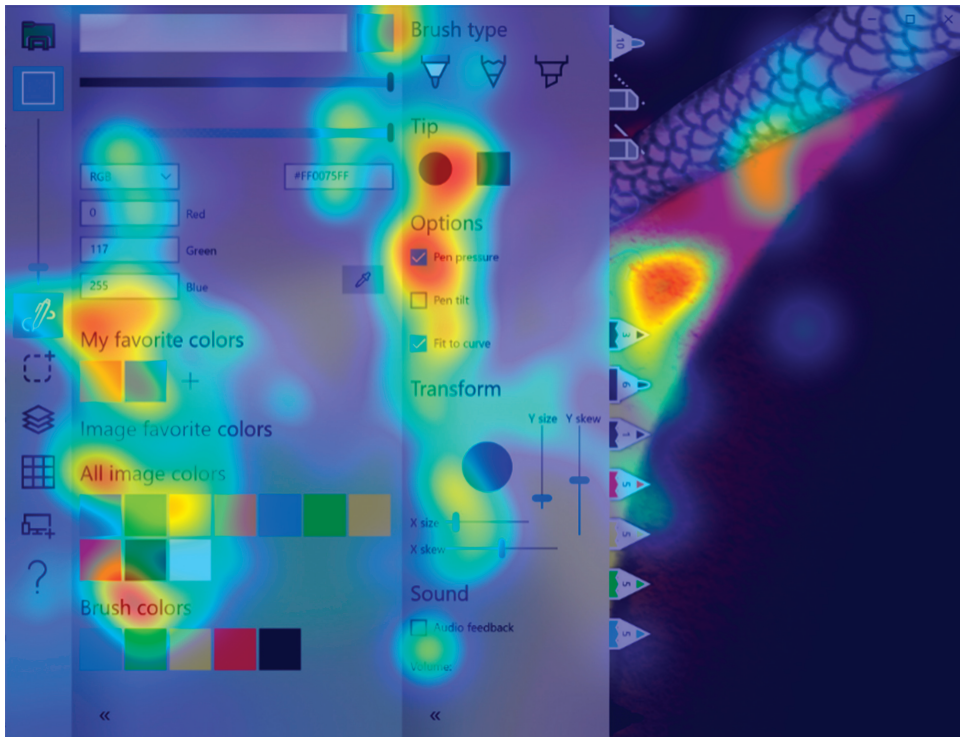


Figure 3.12 This heatmap of eye-tracking data shows what people look at when viewing a UI [385]. Visual saliency denotes the probability with which visual features attract attention when we see a UI for the first time. Saliency depends on the distribution of visual features on the UI, user expectations (including any prior experience with related designs), and the user's attentional strategy.

that are unique in terms of visual primitives, such as color, shape, size, orientation, or motion, stand out.

On the one hand, saliency emerges in the parallel processing of retinal input at lower levels in the visual cortex. On the other hand, visual saliency is also affected by top-down factors such as memory and expectations of informative locations. That is, we learn how to look. For instance, if a color such as red appears in an image full of green tones and green-filled shapes, observers tend to look at the red shape. Top-down factors include task goals and expectations based on the learned statistical distribution of features. For example, in many natural scenes that show a horizon, most of the information lies close to the horizontal medial line, which also attracts attention. When the visual task or content changes, both bottom-up and top-down factors may change in response. Therefore, an empirical effect related to the saliency reported for one context does not carry over to another trivially.

Research has found that there is something special about UIs. They differ from natural scenes (say, a forest) in several respects. With natural scenes, there is a tendency to look at the center and the horizontal line. This is understandable, as the horizontal line often provides the most information about a natural scene. However, for UIs, this is often not the case [472]. Attention tends to be drawn to the top-left quadrant of the page. In addition, color is a poorer predictor of saliency than in natural scenes. Instead, attention is drawn to large texts and images with faces.

However, despite visual saliency having emerged as one of the better-known aspects of the human visual system, designers often use rules of thumb and gut feelings instead. It can also explain what makes a display cluttered (Paper Example 3.3.3.)

Paper Example 3.3.3: What makes a display cluttered?

Take a look at the desk in the picture below (picture used under Creative Commons License). Anyone can see that it is cluttered—but what does clutter mean?

Consider visual clutter as the “evil twin” of saliency. Rosenholtz et al. [712] noted that visual saliency can be used to explain and measure how cluttered something is perceived to be. *Clutter* means that *everything* is attempting to be salient.

The underpinning hypothesis is that the human visual system has evolved to detect unusual objects in scenes. A salient item is unusual in its visual context. Clutter, by contrast, refers to excess items competing for our attention. In the state of clutter, all visual features are “congested.” For example, color clutter means that many colors from the visible spectrum are used in the display. Many elements are unique in this, and, therefore, nothing is salient.

Rosenholtz proposed a simple test to determine if a display is cluttered. If you wanted to place a note on the desk (e.g., a Post-it note) that the user should notice, how would the design be? Which color, size, and shape would you choose? If the desk is cluttered, it may not be possible to pick a visual feature that makes the note stand out on the display.

Clutter is detrimental to visual performance. It is empirically associated with a decrease in object recognition performance and visual search performance, as well as forgetting due to exceeding the limit of short-term memory.



3.4 Perception as an active process

One of the main takeaways from this chapter is that users are not at the mercy of their senses. They are not passive recipients of stimulation determined solely by the display design. Instead, they actively construct percepts.

Users apply their knowledge to actively decide where to focus their attention, including guessing where things are or what they are called. In addition to learning, users strategically change how they perceive things. For example, experienced users rely more on their long-term memory than novices do. The reward–cost structure of a task also affects perception. When a task structure changes, for example, to penalize mistakes and errors, users may change their strategy accordingly.

Moreover, users actively explore their environment using their bodies. They also adjust the relationship their body has with the environment. For example, if you were for some reason abruptly re-positioned one meter farther from this text, you would want to step closer to continue reading it. In short, people use their bodies to facilitate and support their perception of things. This approach is called *active perception*. Active perception refers to perception that is actively sampling, exploring, and learning to better support the organism. *Ecological perception* refers to the idea that these processes are sensitive to and adapt to the structure of the environment. For example, once we have learned to look at web pages in a particular way, we expect items to appear in specific locations that reflect the probability of those items being there. *Embodied perception* refers to a more radical view whereby perception and bodies cannot be decoupled: All perception is neurobiologically coupled to motor systems. No matter the view, it is clear that understanding embodiment is as important for understanding perception as it is for understanding its underpinning sensory modalities.

A prominent example of active perception is *perceived affordance*, a term coined by Norman in the book *The Design of Everyday Things*. Affordance is a property of an element that invites interaction. For example, a door may have an opening mechanism that invites the user to either push a plate or pull a handle to open the door. Users would, in general, know which method is suitable for the door without trying both based on the affordance of the door handle. Graphical objects can also afford interaction, for example, by signaling whether they should be clicked, dragged, or similar. However, Norman argues that the useful property for a GUI is not affordance but perceived affordance: “In similar vein, because I can click anytime I want, it is wrong to argue whether a graphical object on the screen ‘affords clicking.’ It does. The real question is about the perceived affordance: Does the user perceive that clicking on that location is a meaningful, useful action to perform?”²

Buttons, for example, can communicate affordance through shapes, colors, and shadows. However, perceived affordance is cultural. Our ability to perceive affordance in graphical objects relies on our existing knowledge and experience. For example, the blue hyperlinks that are now common did not elicit clicks from users until people learned that it was possible to click on links in the text.

² http://www.jnd.org/dn.mss/affordances_and_design.html.

Summary

- Perceptual experience is not determined by what is shown on the display alone but in conjunction with our expectations and strategies of deploying attention.
- Perception in HCI can be understood through the elementary tasks it needs to serve: discrimination, detection, recognition, estimation, and search.
- Another key role of perception is to “bring order into chaos,” in other words, organize the overwhelming sensory experience into something we can act on.
- Visual saliency refers to the probability with which something on display attracts our visual attention. When the UI is cluttered, all visual primitives are “congested” and nothing can be made to stand out from the page.

Exercises

1. Analysis of a visual design. Choose a GUI you think is difficult to use. Could this be because of the organization of the display? Take a screenshot and analyze it in terms of how it uses (a) the principle of visual saliency (what attracts attention) and (b) the Gestalt laws (grouping of information).
2. Alternative modalities. Section 3.1 lists three types of sensory modalities prevalent in computer use. Consider designing a display that shows the speed of a car to the driver. What are the pros and cons of each sensory modality when used as a display in this case?
3. Perceptual tasks. This chapter has discussed five perceptual tasks in HCI: discrimination, detection, recognition, estimation, and search. Which perceptual task is the defining task in the following cases? (a) Playing a first-person shooter, you see a shadowy character move in the distance; is it a friend or a foe? (b) You arrive at the airport and look at a display of departing flights; from which gate is your flight leaving? (c) You're on the subway and feel a slight vibration in your pocket; is it your mobile device or a tremble from the subway? (d) You're wearing a VR headset and see a button close to you; is it close enough that you can press it without moving, or should you first walk closer to it? (e) You see an icon that has a circle with a cross (X) in the middle; is it a “Close window” button?
4. Using smell in UIs. Compare olfaction (smell) and vision as sensory modalities. Think about everyday HCI: Even if technology were to allow for olfactory displays, why are they unlikely to replace vision-based displays?
5. Understanding clutter. Pick a UI (e.g., a ticket vending machine) that you think looks cluttered. Why is it so? Refer to the conjunction search to identify three regions of the UI that might be hard to find. Circle and number the regions in the UI and explain why they are hard to find.
6. Visual grouping in GUIs. Take a GUI and look at it. Which elements belong together, i.e., are grouped? Which visual grouping principles are applied to achieve this?
7. Unhelpful use of color. A simple mobile device is operated by holding the device with the non-dominant hand and a plastic pen with the dominant hand. The device enables text entry through a graphical keyboard presented on a resistive screen. The system does not perform word prediction or autocorrect. A designer proposes to improve the text entry performance by changing the colors of the keys on the graphical keyboard. Based on the principles taught in this chapter, explain why this approach is fundamentally flawed.